

Common Neurological and Spine Disorders

Learning Objectives

- ⇒ To have a brief understanding about neurological and spine disorders
- ⇒ To know the clinical features of parkinson's disease, stroke, Bell's palsy, motor neuron disease (MND), transverse myelitis, epilepsy (organic)
- ⇒ To know the clinical features of common spine disorders – lumbago sciatica syndrome, brachial neuralgia, cervical spondylosis, lumbar spondylosis
- ⇒ To perform the necessary physical examinations in above mentioned diseases
- ⇒ To order the necessary laboratory investigations and their interpretation

Parkinson's Disease

Parkinson's disease is a idiopathic, slowly progressive, degenerative disease of central nervous system (CNS) characterized by resting tremor, muscular rigidity, slow and hypokinesia (slow and decreased movement) and postural instability.

Diagnosis done by clinical evaluation. **Parkinsonism**¹ refers to the symptoms that are similar to those of Parkinson's disease, but caused by another condition.

- ⊗ This disease affects the individuals with age of ≥ 65 years. Rarely, this disease afflicts the children and adolescents.
- ⊗ The disease is produced due to the defect observed in the substantia nigra of the midbrain where in depletion in the production of dopamine is present.
- ⊗ This disease is characterized by the presence of symptom triads – resting tremor, bradykinesia or hypokinesia and **muscular rigidity**².



Figure 167: Parkinson's disease

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- 1 Parkinsonism may result from adverse drug reactions, disorders other than Parkinson's disease such as amyotrophic lateral sclerosis (ALS), cerebrovascular disease (CVA), brain tumours present near basal ganglia, repeated traumatic brain injury, hydrocephalus, hypoparathyroidism, methyldopa overdose, reserpine adverse drug reaction, carbon monoxide poisoning, methanol poisoning, manganese poisoning.
 - 2 Assessment of rigidity comes in examination of nervous system. This is coming under motor system examination. The term rigidity refers to increased tone of the muscles which is seen throughout the movement of the limb. Spasticity is another abnormal muscle tone which is seen for a short period of time, not throughout the movement.
 - Rigidity is of two types. Cogwheel rigidity and lead pipe rigidity. The term cogwheel refers to the movement that diminishes in jerky steps due to tremor as well as rigidity. Lead pipe rigidity is observed without tremor.
 - Cogwheel rigidity is seen in upper limbs and lead pipe rigidity is seen in lower limbs of the patients of Parkinson's disease.

- **Resting tremor:** slow and coarse tremor, maximal at rest, lessening during movement and almost absent during sleep; amplitude increased by emotional tension or fatigue; often involving the wrist and fingers in movements similar to those used to manipulate small objects or pills (pill rolling movements); usually hands, arms and legs are most affected in that order. The jaw and tongue may also be affected, but not the voice. Tremor may become less prominent, as the disease progresses.
- **Muscular rigidity:** This may develop without tremor in many patients. When a clinician moves a rigid joint, sudden, rhythmic jerks due to variations in the intensity of the rigidity occur, producing a ratchet like effect (cogwheel rigidity).
- **Bradykinesia (hypokinesia):** The patient may have slower movements as rigidity progresses. Movement also becomes decreased and later it becomes difficult to initiate the movement (akinesia).
- ⊙ Rigidity and hypokinesia may contribute to muscular aches and sensations of fatigue. The individual suffering from this disease will have expressionless face or masklike face with open mouth, drooling and reduced blinking of the eyes. Early on the patient may appear depressed because of lack of facial expression and slowed and reduced movements of the body. Speech becomes hypophonic, with characteristic monotonous, stuttering dysarthria.
- ⊙ Hypokinesia and impaired control of distal musculature leads to micrographia (writing in very small letters) and make the daily activities very difficult. Without warning, voluntary movements of the patient including walking may suddenly halt (freezing).
- ⊙ Postural instability develops, resulting in gait abnormalities which is termed as short shuffling gait (Parkinsonian gait or festinating gait).
- ⊙ In this, patient will have difficulty in starting the walking, burning and stopping; the becomes shuffling with short steps, and arms are flexed to the waist and do not swing with the stride. Steps may inadvertently quicken, and patients may break into a run to keep from falling (festination).
- ⊙ A tendency to fall forward (propulsion) or backward (retropulsion) when the centre of the gravity is displaced results from the loss of postural instability. Posture becomes stooped.
- ⊙ Dementia may occur.
- ⊙ Diagnosis is clinical. This disease is suspected in patients with characteristic unilateral resting tremor, decreased movement or rigidity.

Stroke

The term stroke is also known as cerebrovascular accident (CVA). These disorders are heterogenous group of disorders involving sudden, focal interruption of cerebral blood flow that causes neurological deficit. Ischaemia in the cerebral region (ischaemic stroke in 85% of the patients) and intracranial haemorrhage (haemorrhagic stroke in 15% of the patients) are the two mechanisms by which CVA manifests in the individuals. This disease has to be differentiated from seizures, intracranial tumour, migraine and metabolic encephalopathy. Symptoms observed in the patient depends upon the site of the lesion, size of the lesion. Abrupt onset of numbness, weakness or paralysis of the contralateral limbs and face; aphasia; confusion are observed during the progression of the disease. Diagnosis is done by CT scan of the head. CBC, lipid profile, ECG etc will help in predicting the prognosis of the disease.

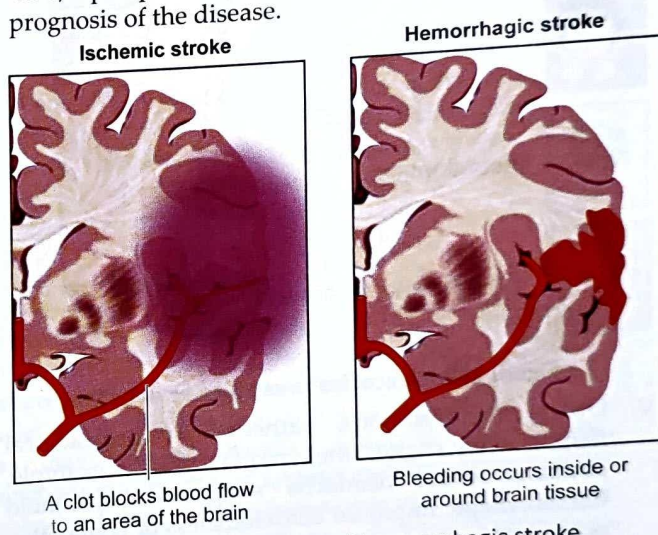


Figure 168: Ischemic and haemorrhagic stroke

- ⊙ Previous history of cardiac disease or carotid atherosclerosis or intracerebral aneurysms are the main reasons for the development of CVA. Prior history of stroke (TIA transient ischemic attack); older age; family history of stroke; alcoholism; male gender; hypertension; cigarette smoking; hypercholesterolemia; diabetes mellitus and use of certain drugs such as cocaine or amphetamines) are some of the risk factors for the development of stroke (CVA).
- ⊙ Initial symptoms (see Figure 169) occur suddenly. Generally, they include numbness, weakness or paralysis of the contralateral limbs and face; aphasia; confusion; visual disturbances in one or both eyes.
- ⊙ Neurological deficits reflect the area of the brain involved. Anterior circulation stroke typically causes unilateral symptoms. Posterior circulation stroke can cause unilateral or bilateral deficits and is more likely to affect consciousness, especially when the basilar artery is involved,

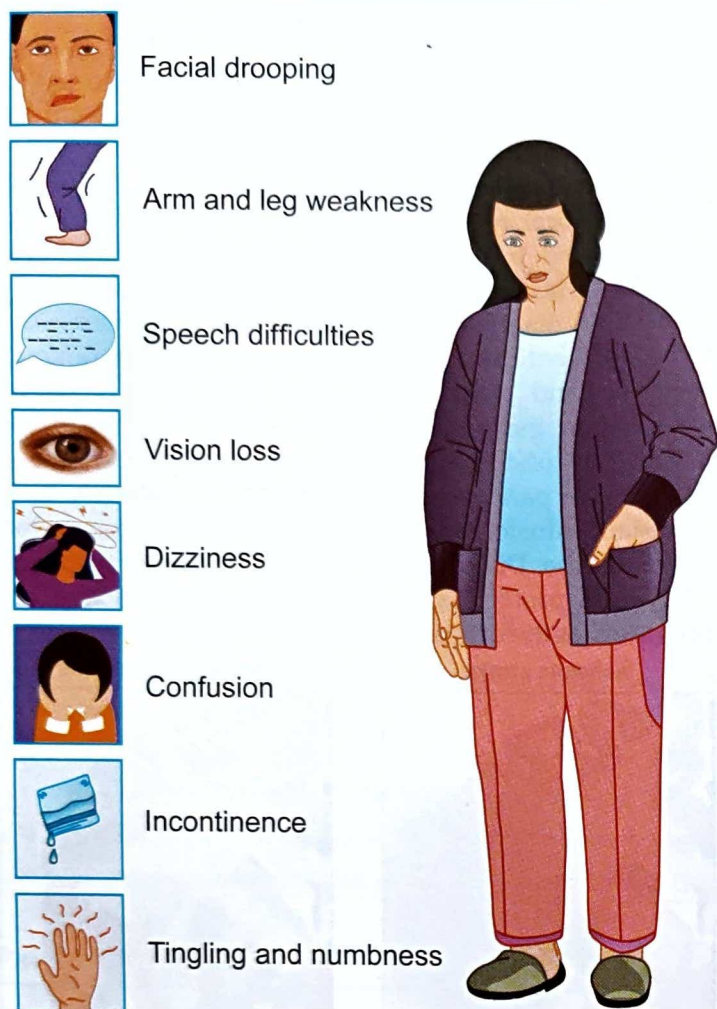


Figure 169: Clinical features of stroke or CVA

- ❖ Other manifestations, rather than neurological deficits, often suggest the type of stroke. For example, sudden, severe headache suggest subarachnoid haemorrhage. Impaired consciousness or coma often accompanied by headache, nausea, vomiting suggest increased intracranial pressure which can occur 48 to 72 hours after large ischaemic strokes.
- ❖ Complications of stroke (CVA) can include sleep disturbances; confusion; depression; incontinence; atelectasis; pneumonia and swallowing dysfunction which can lead to aspiration, dehydration or undernutrition. Immobility can lead to deep vein thrombosis in the lower limbs or pressure sores in the back or electrolyte imbalances.
- ❖ Based on the location of the affliction of artery, clinical symptoms may vary.
 - **MCA (Middle cerebral artery) territory infarct:** The patient may show contralateral hemiplegia, hemianaesthesia with or without homonymous hemianopia and aphasia. Occlusion of upper division results in contralateral hemiparesis with sensory deficit and expressive aphasia (Broca's aphasia), whereas Wernicke's aphasia is commonly seen with occlusion of lower division on the dominant side.

- **ACA (Anterior cerebral artery) territory infarct:** Clinical features include sensory motor paralysis of the opposite lower extremity, with mild weakness of the opposite shoulder. Mental changes, rectal and urinary incontinence, gait disturbances, apraxia may be observed. Aphasia and hemianopia are never seen.
- **PCA (Posterior cerebral artery) territory infarct:** Contralateral homonymous hemianopia is a significant finding and this results from infarction of the primary visual area, the central vision is frequently spared. Distorted vision, visual object agnosia, various forms of dyslexia and dysgraphia are being observed.

❖ Diagnosis is done by CT scan (see Figure 170a) head to determine the cause of cerebrovascular accident either for infarction or haemorrhage (see Figure 170b). Prognosis is determined by CBC, ESR, lipid levels and urine routine analysis.



Figure 170 a & b: Diagnosis of ischemic and haemorrhagic stroke respectively

Bell's Palsy

Bell's palsy is sudden, idiopathic, unilateral peripheral 7th cranial nerve palsy. Symptoms are hemifacial paralysis of the upper and lower face. There are no specific tests for the diagnosis of the disease.

- ⊗ Cause is unknown, but the mechanism is presumably swelling of the 7th cranial nerve due to an immune or viral disorder. Herpes simplex infection may be the triggering factor for the manifestation of the disease.

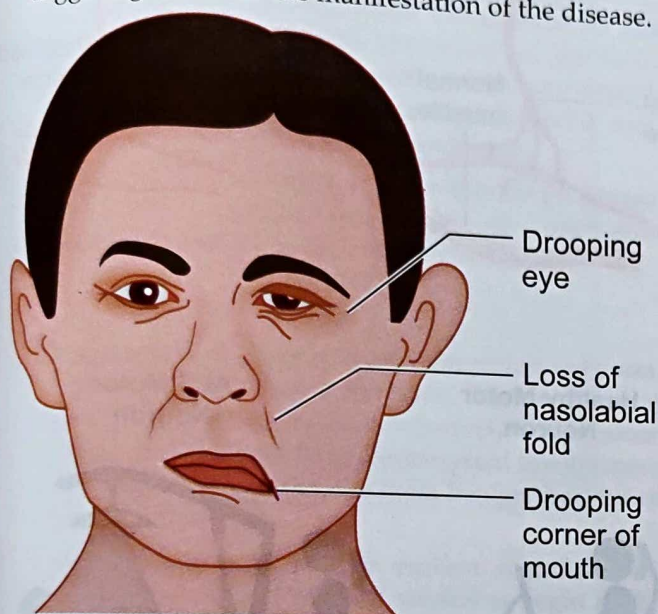


Figure 171: Bell's palsy

- ⊗ Pain behind the ear precedes the facial paresis. Paresis, often with complete paralysis develops within 24 hours and usually maximal within 48 to 72 hours. Patients may have numb or heavy feeling in the face. The affected side becomes flat and expressionless; ability to wrinkle the forehead, blink and grimace is limited or absent. In severe cases, palpebral fissure

widens and the eye does not close, often irritating the conjunctiva.

- ⊗ Sensory examination is normal, but the external auditory canal and a small patch behind the ear (over mastoid region) may be painful to touch. If the nerve lesion is proximal to the geniculate ganglion, excess salivation, taste and lacrimation may get impaired.
- ⊗ Diagnosis is by clinical evaluation. Middle ear or mastoid infection has to be excluded. Other diseases to be excluded include sarcoidosis, petrous bone fractures, chronic meningitis, cerebellopontine tumours. Other disorders may show the clinical symptoms slowly when compared to the features of Bell's palsy.
- ⊗ Prognosis of Bell's palsy is determined by the extent of nerve damage. If some function of facial nerve remains, full recovery typically occurs within several months.
- ⊗ Nerve conduction studies and electromyography may help in predict the outcome of the disease. The likelihood of complete recovery after total paralysis is 90% if nerve branches in the face retain normal excitability. Chronic disuse of the facial muscles may lead to contractures.

Motor Neuron Disease

Motor neuron disease (see Figure 172) are characterized by steady, relentless, progressive degeneration of corticospinal tracts, anterior horn cells, bulbar motor nuclei or a combination. Symptoms vary in severity and may include muscle weakness and atrophy, fasciculations, emotional lability and respiratory muscle weakness. Diagnosis involves nerve conduction studies, electromyography and exclusion of other disorders via MRI and laboratory investigations.

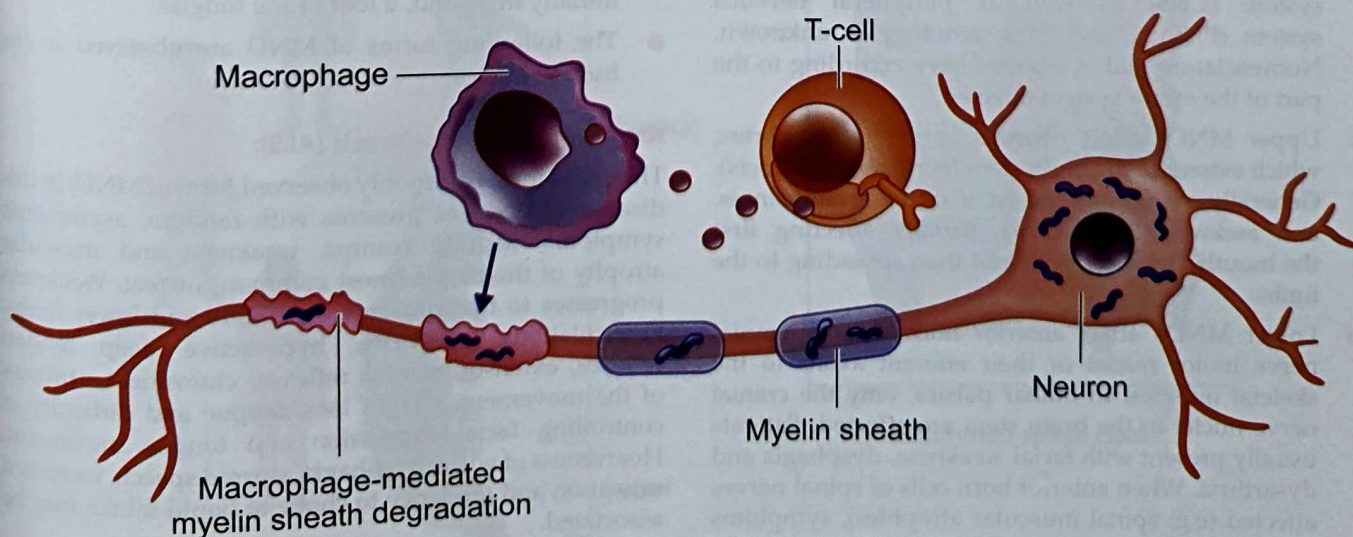


Figure 172: Pathology of Motor Neuron Disease

Amyotrophic Lateral Sclerosis (ALS)

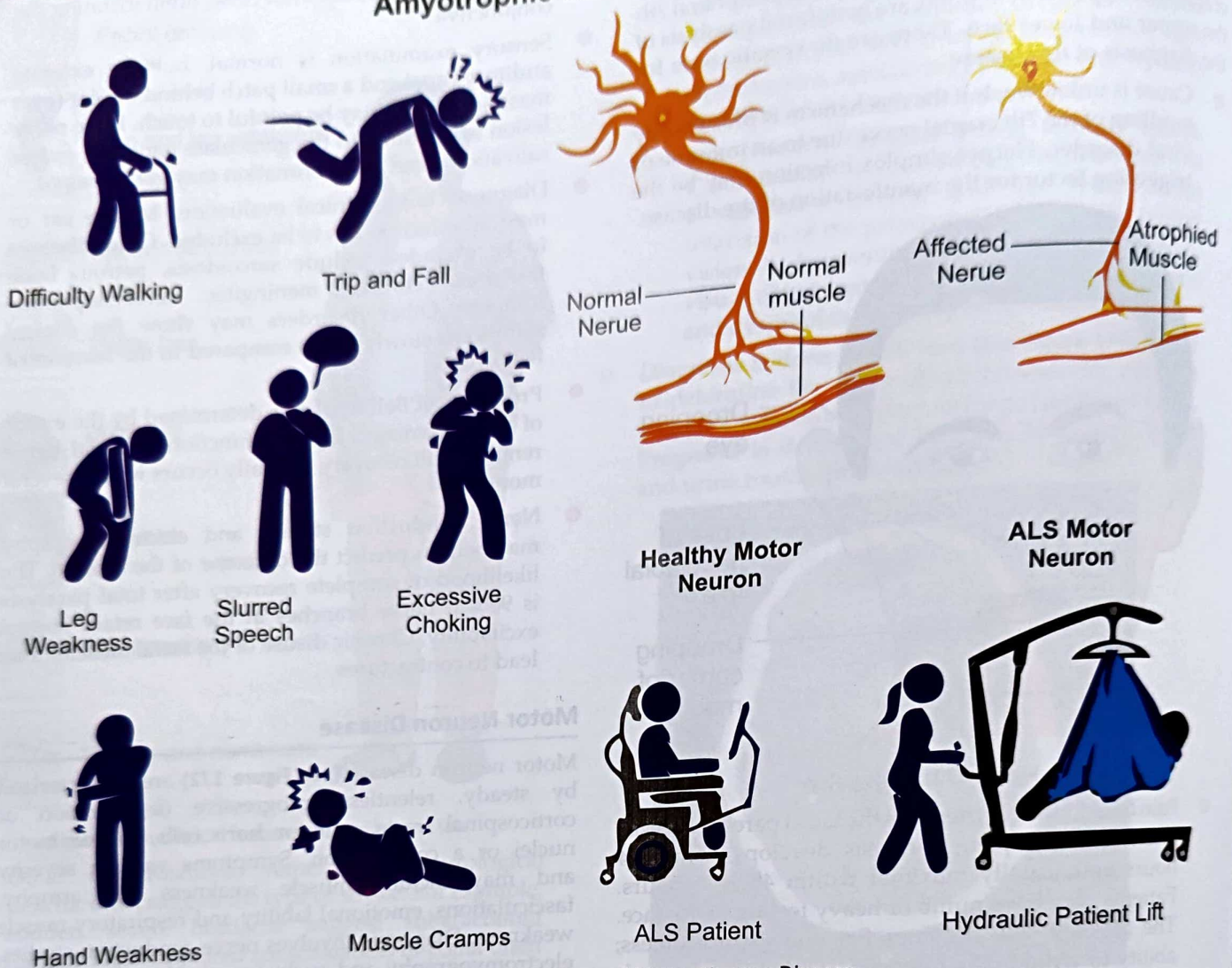


Figure 173: Clinical features of Motor Neuron Disease

- MNDs (see Figure 173) may involve central nervous system (CNS) as well as peripheral nervous system (PNS). Usually the aetiology is unknown. Nomenclature and symptoms vary according to the part of the motor system affected.
- Upper MNDs affect neurons of the motor cortex, which extend to the brain stem (corticobulbar tracts). Generally, symptoms consist of stiffness, clumsiness, and awkward movements, usually affecting first the mouth, throat or both and then spreading to the limbs.
- Lower MNDs affect anterior horn cells or cranial nerve motor nuclei or their efferent axons to the skeletal muscles. In bulbar palsies, only the cranial nerve nuclei in the brain stem are affected. Patients usually present with facial weakness, dysphagia and dysarthria. When anterior horn cells of spinal nerves affected (e.g. spinal muscular atrophies), symptoms include muscle weakness and atrophy, fasciculations

(visible muscular twitches), and muscle cramps, initially in a hand, a foot or the tongue.

- The following forms of MND are observed in the human beings.

Amyotrophic lateral sclerosis (ALS):

This is the most commonly observed form of MND. In this disease, the patient presents with random, asymmetric symptoms, include cramps, weakness and muscular atrophy of the hands (most common) or feet. Weakness progresses to the forearms, shoulders and lower limbs. Fasciculations, spasticity, hyperactive deep tendon reflexes, extensor plantar reflexes, clumsiness, stiffness of the movement, weight loss, fatigue and difficulty in controlling facial expression and tongue movement. Hoarseness of voice, dysphagia, slurred speech, increased salivation and tendency to choke to liquid intake may be associated.

- Death is caused by failure of respiratory muscles;

50% of the patients die within 3 years of onset; 20% of the patients will have life span for 5 years and 10% of the patients will have life span of 10 years. Survival of the patient for > 30 years is rare.

Progressive bulbar palsy

- The muscles innervated by cranial nerves and corticobulbar tracts are predominantly affected, causing progressive difficulty in chewing, swallowing, and talking, nasal voice; reduced gag reflex; fasciculations, weak movement of the facial muscles and tongue; and weak palatal movement. A pseudobulbar effect, with emotional lability may occur if the corticobulbar tract is affected.
- Patients with dysphagia will have poor prognosis. Respiratory complications due to aspiration frequently result in death within 1 to 3 years.

Progressive muscular atrophy

- In many cases, especially those with childhood onset, inheritance is autosomal recessive. This can occur at any age. Anterior horn cell involvement occurs alone or is more prominent than corticospinal involvement and progression tends to be more benign than that of other MNDs.
- Fasciculations may be the earliest manifestation. Muscle wasting and marked weakness begin in the

hands and progress to the arms, shoulder and legs, eventually become generalized. Patients may survive ≥ 25 years.

Primary lateral sclerosis and progressive pseudobulbar palsy

- Muscle weakness and signs of distal motor weakness gradually increase, affecting the limbs in primary lateral sclerosis and lower cranial nerves in progressive pseudobulbar palsy. Fasciculations and muscle atrophy may follow many years later.
- Diagnosis is done by electrodiagnostic tests, MRI of brain laboratory investigations to rule out other diseases. Tests include CBC, electrolytes, creatinine kinase, thyroid function tests to rule out other clinical conditions.

Transverse Myelitis

Transverse myelitis (see Figure 174) is acute inflammation of gray and white matter in one or more adjacent spinal cord segments, usually thoracic. Causes include multiple sclerosis, infections, autoimmune or postinfectious inflammation, vasculitis and certain drugs. Symptoms include bilateral motor, sensory and sphincter deficits below the level of the lesion. Diagnosis is done by MRI, CSF analysis and blood tests.

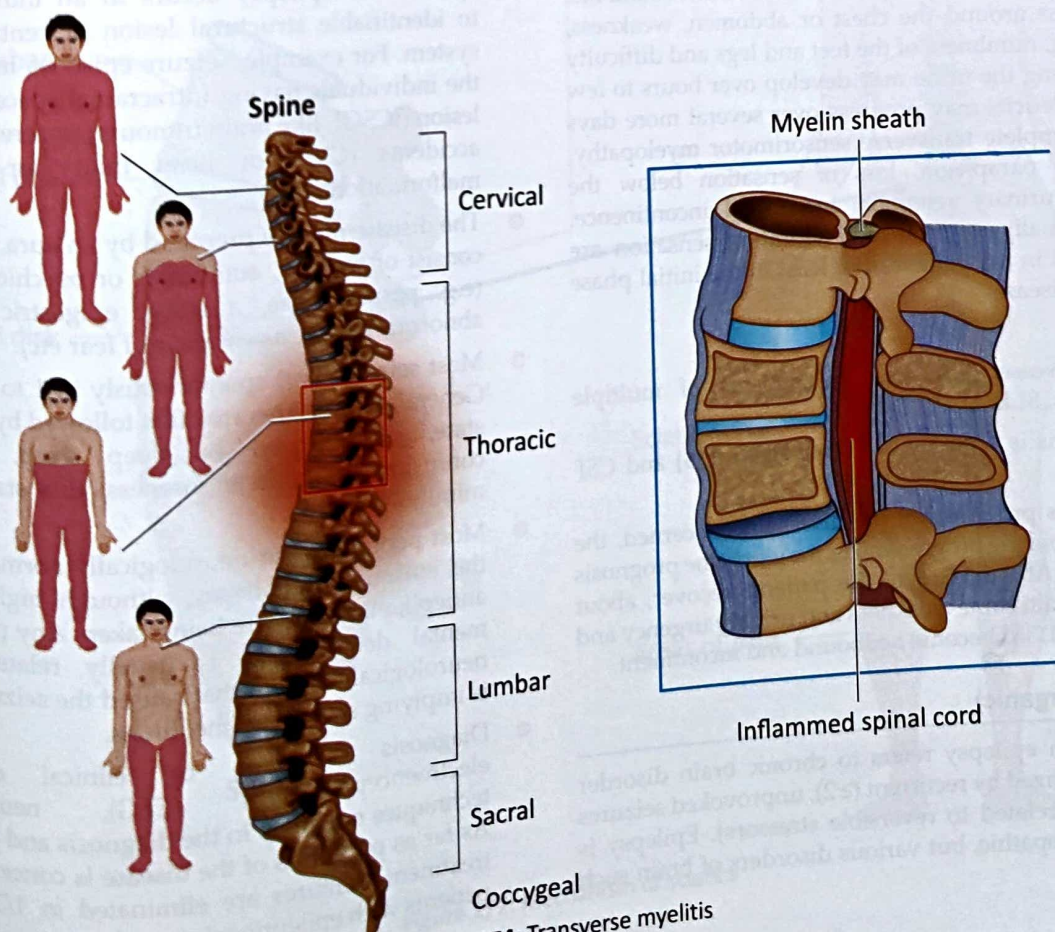


Figure 174: Transverse myelitis



Figure 175: Diagnosis of transverse myelitis

- ✿ Probable causes include viral meningitis, tuberculosis, IV heroin administration or antiparasitic drugs or antifungal drugs, multiple sclerosis or vasculitis.
- ✿ Pain in the neck, back or head may occur. A band like tightness around the chest or abdomen, weakness, tingling, numbness of the feet and legs and difficulty in voiding the urine may develop over hours to few days. Deficits may progress over several more days to a complete transverse sensorimotor myelopathy, causing paraplegia, loss of sensation below the lesion, urinary retention and faecal incontinence. Occasionally, position and vibration sensation are retained in the patient or at least in the initial phase of the disease.
- ✿ This disease recurs in the patients of multiple sclerosis, SLE.
- ✿ Diagnosis is done by MRI (see **Figure 175**) and CSF analysis.
- ✿ As far as prognosis of the disease is concerned, the more rapid the progression is, the worse the prognosis will be. About 1/3rd of the patients recover, about 1/3rd retain some weakness and urinary urgency and another 1/3rd become bedbound and incontinent.

Epilepsy (Organic)

- ✿ The term epilepsy refers to chronic brain disorder characterized by recurrent (≥ 2), unprovoked seizures (i.e. not related to reversible stressors). Epilepsy is often idiopathic, but various disorders of brain such

as malformations, cerebrovascular accidents (CVA) or brain tumours can cause symptomatic epilepsy. Seizure is synonymously used for the epilepsy. It is defined as an abnormal, unregulated electrical discharge that occurs within the brain's cortical gray matter and transiently interrupts motor brain function.



Figure 176: CT scan of organic epilepsy

- ✿ The term organic epilepsy refers to episode or episodes of epilepsy occurs in an individual due to identifiable structural lesion of central nervous system. For example, seizure episodes may occur in the individuals having intracranial space occupying lesion (ICSOL like brain tumours) or cerebrovascular accidents (CVA) or head injury or congenital malformations etc.
- ✿ The disease may be preceded by an aura. Auras may consist of sensory, autonomic or psychic sensations (e.g. paraesthesias, a rising epigastric sensation, abnormal smells, a sensation of fear etc).
- ✿ Most seizures end spontaneously in 1 to 2 minutes. Generalized seizures are often followed by a postictal state, characterized by deep sleep, headache, confusion and muscle soreness; this state last for minutes to hours.
- ✿ Most people appear neurologically normal between the episodes of epilepsy, although high doses of antiepileptic drugs are being taken. Any progressive mental deterioration is usually related to the neurological disorder that caused the seizures which is implying severity of the illness.
- ✿ Diagnosis is done by clinical evaluation; electroencephalogram (EEG), neuroimaging techniques may help in the diagnosis and prognosis. As far as prognosis of the disease is concerned, with treatment, seizures are eliminated in 1/3rd of the patients with epileptic seizures; frequency of seizures

is reduced by 50% in another 1/3rd of the patients. About 60% of patients whose seizures are well controlled by drugs can eventually stop the drugs and remain seizure free. Sudden unexplained death in epilepsy (SUDEP) is rare complication of unknown case of epilepsy.

Lumbago Sciatica Syndrome

The term lumbago refers to lower back pain. The term sciatica (see Figure 177) refers to the pain along the sciatic nerve. It usually results from compression of nerve roots in the lower back. Common causes include intervertebral disc herniation, osteophytes and narrowing of spinal canal (spinal stenosis). Symptoms include pain radiating from lower back or buttocks to the lower limb. Diagnosis sometimes, involves MRI or CT scan.

- Sciatica is typically caused by nerve root compression, usually due to intervertebral disc herniation, bony irregularities or intraspinal tumours or abscesses. Compression may occur within the spinal canal or

intervertebral foramen.

- Pain radiates along the course of the sciatic nerve, most often down the buttocks and posterior aspect of the leg to below the knee. The pain is typically burning, lancinating or stabbing. It may occur with or without low back pain. Coughing, sneezing, straining for defaecation or Valsalva maneuver may worsen the back pain due to disc herniation. The patient may also complain numbness and sometimes weakness in the affected lower limb.
- Nerve root compression can cause sensory, motor signs or reflex deficits (L5-S1 disc herniation may affect ankle jerk, L3-L4 disc herniation may affect knee jerk). Straight leg raising (SLR) test may cause pain that radiates down the leg when the leg is raised above 45°. Crossed SLR test is more specific for the diagnosis of sciatica.
- Diagnosis is by clinical evaluation and sometimes by MRI or CT scan. Sometimes, electrodiagnostic studies may help in the diagnosis.

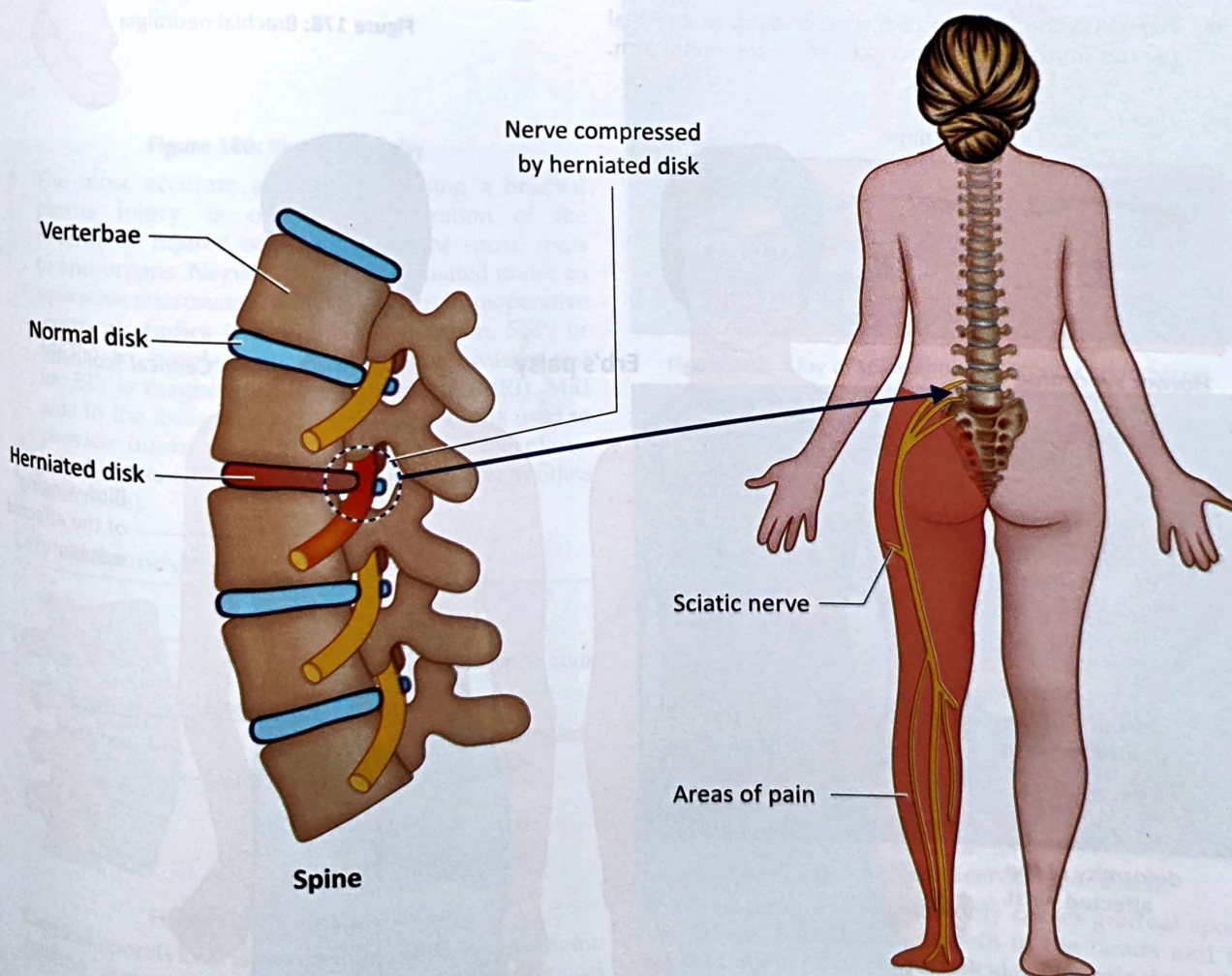


Figure 177: Mechanism of sciatica

Brachial Neuralgia

The term brachial neuralgia refers to a characteristic clinical syndrome of abrupt, severe pain in the shoulder and upper arm, that is followed several weeks later by muscle weakness and profound muscle wasting. It is also known as brachial neuritis, brachial amyotrophy. This is often associated with preceding viral illness, vaccination or other immunological event. Affected muscles are most commonly supplied from C5/C6 segmental level, suggesting the involvement of upper trunk of brachial plexus, such as the long thoracic, axillary, suprascapular and anterior interosseus nerves.

- The condition is said to be manifested due to idiopathic aetiology or occurs as a result of shoulder injury or occurs as a result of fall from a height onto the side of the head or shoulder or as a result of shoulder dystopia in case of obstructed labour.
- **Clinical features** (see Figure 178) may include a limp or paralysed arm, lack of muscle control in the arm, hand and lack of feeling in the arm or hand.
- Based on the location of the nerve damage, brachial plexus injuries can affect part of or the entire arm.

E.g. musculocutaneous nerve damage weakens elbow flexors, median nerve damage causes forearm pain and paralysis of ulnar nerve causes weak grip and finger numbness. The cardinal signs of brachial plexus injury will have weakness in the arm, diminished reflexes and corresponding sensory deficits.



Figure 178: Brachial neuralgia

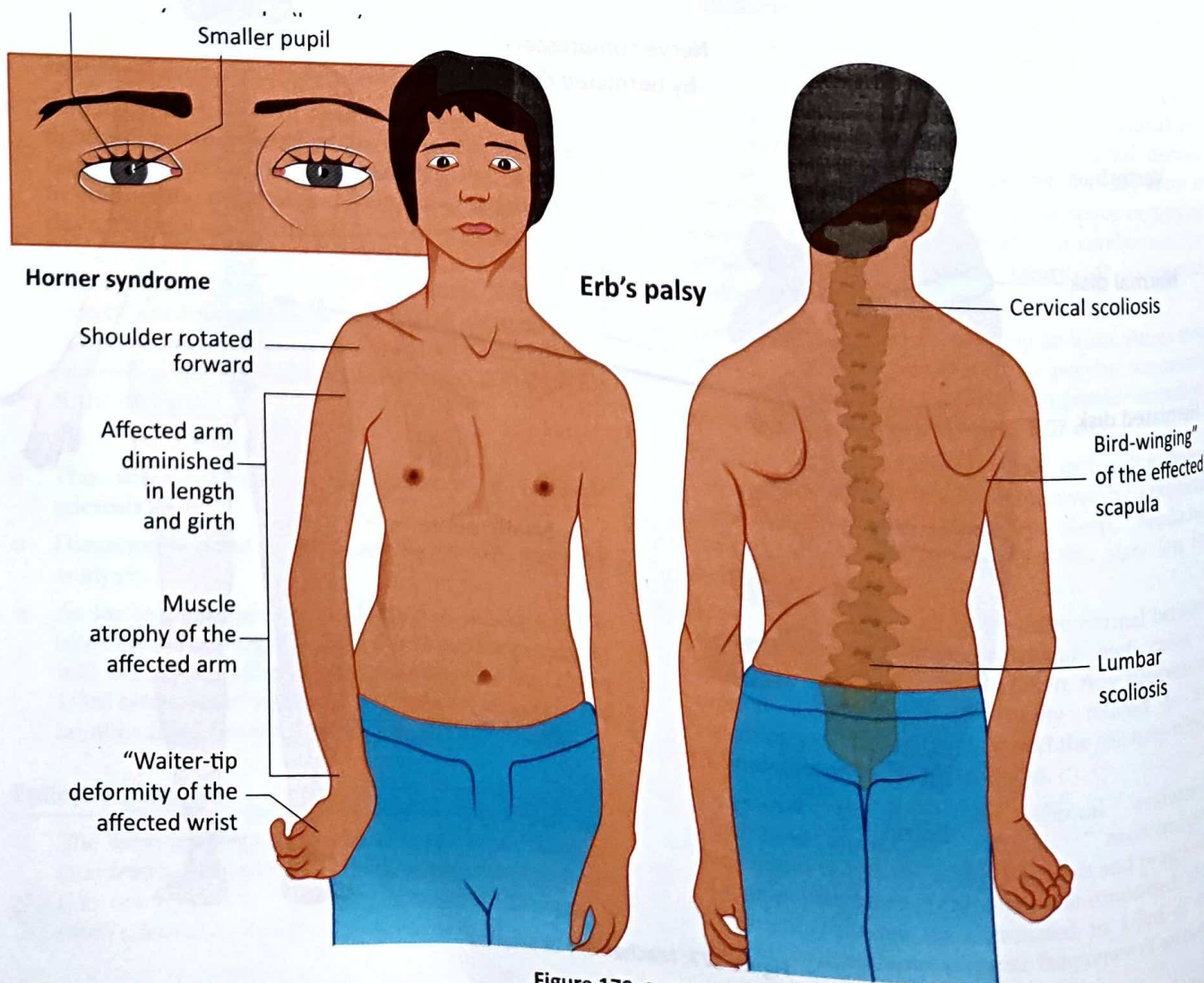


Figure 179: Erb's palsy

➤ **Erb's palsy:** (see Figure 179) The position of the limb, under such conditions is characteristic: the arm hangs by the side and is rotated medially; the forearm is extended and pronated. The arm cannot be raised from the side; all power of flexion of the elbow is lost, as is also supination of the forearm.

➤ **Klumpke's paralysis:** (see Figure 180) A form of paralysis involving the muscles of the forearm and hand, a characteristic sign is the clawed hand, due to loss of function of the ulnar nerve and the intrinsic muscles of the hand it supplies.

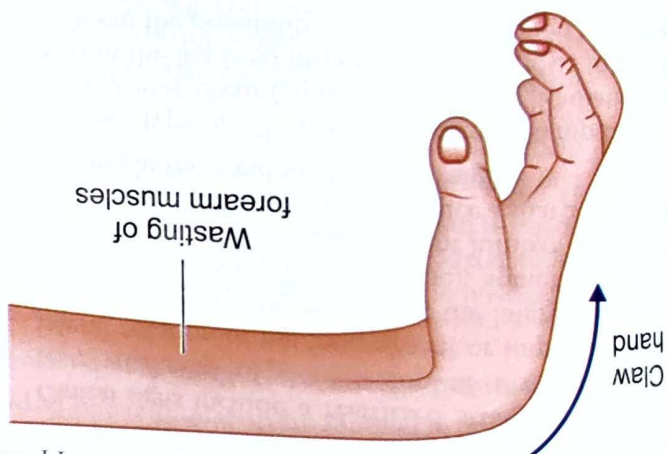


Figure 180: Klumpke's palsy

➤ The most accurate test for diagnosing a brachial plexus injury is operative exploration of the potentially injured segments from the spinal roots to end-organs. Nerves should be evaluated under an operative microscope, with or without intraoperative electrical studies (e.g. bipolar stimulation, SEP or MEPs) to supplement. The best non-invasive test for BPI is magnetic resonance imaging (MRI). MRI aids in the assessment of the injuries and is used to provide information on the portion of the plexus which cannot be operatively explored (the rootlets and roots).

Cervical Spondylosis

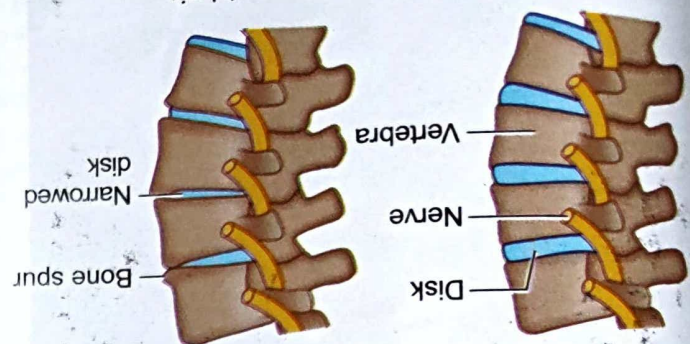


Figure 181: Cervical spondylosis

Cervical spondylosis is osteoarthritis of the cervical spine causing stenosis of the canal and sometimes cervical

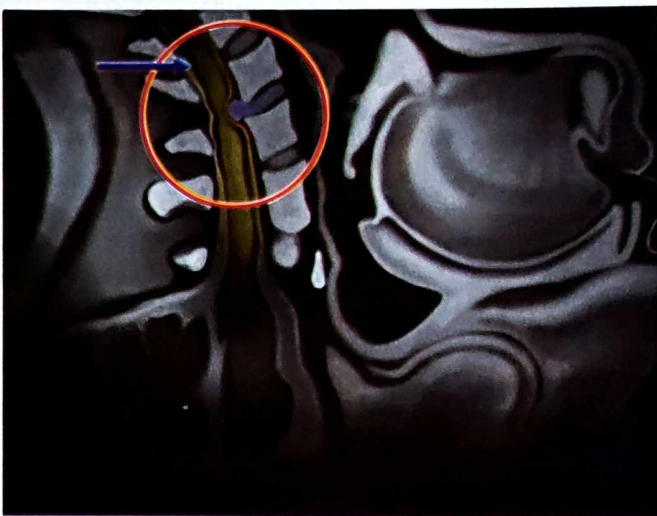
➤ **Cervical spinal canal is congenitally narrow in some roots (radiculomyelopathy).** Sometimes with involvement of lower cervical nerve growths (osteophytes) on the lower cervical spinal cord, myelopathy due to encroachment of bony osteoarthritic roots (radiculomyelopathy). This may occur as a result of the inflammation of the cervical spine. This may lead to the compression of spinal cord or spinal nerves leading to the symptoms of intervertebral disc prolapse in the cervical region.



Figure 182: X ray of spine showing cervical spondylosis

➤ **Cord compression commonly causes gradual spastic paresis, paresthesias or both in the hands and feet and may cause hyperreflexia. Neurologic deficits**

Figure 183: MRI scan of cervical spondylosis



may be asymmetric, nonsegmental, and aggravated by coughing, sneezing and straining for defaecation or any maneuvers which increase the intraspinal pressure. Eventually, the person may develop muscle atrophy and flaccid paresis in the upper limbs at the level of the lesion, with spasticity below the level of the lesion.

- ⊗ Nerve root compression commonly causes early radicular pain, later may be weakness, hyporeflexia and muscle atrophy.
- ⊗ Diagnosis (see Figure 182) is done by MRI (see Figure 183) or CT scan. Cervical spondylosis is suspected when characteristic neurological deficits occur in patients who are elderly, have osteoarthritis or have radicular pain at the level of C5 or C6 vertebra.

Lumbar Spondylosis

The term lumbar spondylosis (see Figure 184) is a degenerative disorder of the lumbo-sacral spine giving rise to disc degeneration and osteophytes formation, leading to spinal deformity, low back pain radiating down one or both lower limbs and features of lumbar radiculopathy.

- ⊗ The pathogenesis of lumbar spondylosis is similar to that of the cervical region, there are two anatomical differences relevant to the former's pathophysiology. First, the spinal cord ends opposite to the lower border of L1 vertebra. Therefore, lumbar disc lesion can only cause root syndromes (radiculopathy) and not cord compression syndrome (compressive myelopathy) and hence spastic paraplegia. As there are 8 cervical roots and 7 cervical vertebrae, the relationship of the root to the interspace alters only below T1. A disc lesion at L4/L5 damages L5 root and a disc lesion at L5/S1 damages the S1 root (the commonest lesion in lumbar spondylosis).
- ⊗ The actual cause of back and leg pain is compression of the nerve root, disc diseases leading to herniated disc is the principal factor.
- ⊗ Lumbar canal stenosis, both congenital and acquired, is another important factor. Other causes of pain such as facet arthropathies and hypertrophy of posterior longitudinal ligaments as occurring in cervical spondylosis.
- ⊗ The most common presenting feature in the middle aged is the so called sciatica syndrome, due to protrusion of the degenerated L5/S1 disc which produces acute radiculopathy.
- ⊗ There is a sudden, sharp, low-back pain (LBP) with a feeling of something having snapped at the back. The pain occurs during active movement, stooping or lifting a weight.
- ⊗ It radiates along either leg down to the great toe; it is worse with coughing and bending and is terrible during an unguarded sneeze.

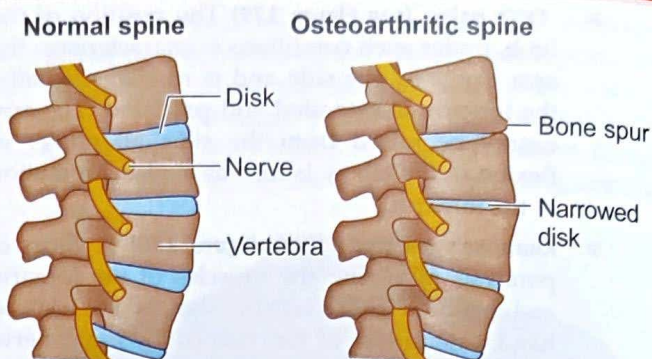


Figure 184: Lumbar spondylosis

- ⊗ Clinical signs include a restricted straight-leg-raising test, weakness of dorsiflexion of the hallux and lateral rotation of the foot, absent or impaired ankle jerk and sensory impairment at the lateral aspect of the affected foot (S1 dermatome); sometimes there may be tenderness of the relevant lumbar spine and pressure in that region may produce a sharp pain and tingling at the great toe (doorbell sign).
- ⊗ Another type of LBP without radicular symptoms in the legs may occur (so-called lumbago). Radiation of pain in the leg is so important in diagnosis that if it is absent the possibility of a disc protrusion is hardly entertained.



Figure 185: X ray spine showing lumbar spondylosis

- Occasionally, a lesion at L3/4 level may produce severe LBP without radicular pain. The most ominous historical feature is rest pain.
- The third type of presentation is the so-called cauda equina syndrome.
- On a chronically spondylotic spine, an acute disc prolapse at L2/3 or L3/4 may lead to bilateral multiple root lesions, with or without unilateral predominance. There is severe bilateral leg pain, flaccid paraplegia and retention of urine.
- On sensory testing there is appropriate sensory impairment and loss/impairment of knee jerk. Distension of bladder due to retention may be present.
- In advanced chronic cases, there may be wasting of muscles supplied by the affected roots, and bladder disturbances.
- Lumbar facet joints are an accepted generator of pain. Characteristically, pain is felt on extension of the spine.
- Diagnosis is done by MRI (see Figure 185) or CT scan. The other conditions giving rise to LBP include spinal tumour, primary or secondary, including blood dyscrasias; pelvic causes of compression of lumbar plexus; and various forms of arachnoiditis.
- Functional LBP with radiological changes needs careful consideration, consistent with age, sex, personality traits and other stress factors leading to neurotic behaviour pattern.

Key summary of the chapter

- Parkinson's disease is a idiopathic, slowly progressive, degenerative disease of central nervous system (CNS) characterized by resting tremor, muscular rigidity, slow and hypokinesia (slow and decreased movement) and postural instability. Diagnosis done by clinical evaluation.
- The term stroke is also known as cerebrovascular accident (CVA). These disorders are heterogenous group of disorders involving sudden, focal interruption of cerebral blood flow that causes neurological deficit. Ischaemia in the cerebral region (ischaemic stroke in 85% of the patients) and intracranial haemorrhage (haemorrhagic stroke in 15% of the patients) are the two mechanisms by which CVA manifests in the individuals. This disease has to be differentiated from seizures, intracranial tumour, migraine and metabolic encephalopathy. Symptoms observed in the patient depends upon the site of the lesion, size of the lesion. Abrupt onset of numbness, weakness or paralysis of the contralateral limbs and face; aphasia; confusion are observed during the progression of the disease. Diagnosis is done by CT scan of the head. CBC, lipid profile, ECG etc will help in predicting the prognosis of the disease.
- Bell's palsy is sudden, idiopathic, unilateral peripheral 7th cranial nerve palsy. Symptoms are hemifacial paralysis of the upper and lower face. There are no specific tests for the diagnosis of the disease.
- Motor neuron disease are characterized by steady, relentless, progressive degeneration of corticospinal tracts, anterior horn cells, bulbar motor nuclei or a combination. Symptoms vary in severity and may include muscle weakness and atrophy, fasciculations, emotional lability and respiratory muscle weakness. Diagnosis involves nerve conduction studies, electromyography and exclusion of other disorders via MRI and laboratory investigations.
- Transverse myelitis is acute inflammation of gray and white matter in one or more adjacent spinal cord segments, usually thoracic. Causes include multiple sclerosis, infections, autoimmune or postinfectious inflammation, vasculitis and certain drugs. Symptoms include bilateral motor, sensory and sphincter deficits below the level of the lesion. Diagnosis is done by MRI, CSF analysis and blood tests.
- The term epilepsy refers to chronic brain disorder characterized by recurrent (≥ 2), unprovoked seizures (i.e. not related to reversible stressors). Epilepsy is often idiopathic, but various disorders of brain such as malformations, cerebrovascular accidents (CVA) or brain tumours can cause symptomatic epilepsy. Seizure is synonymously used for the epilepsy. It is defined as an abnormal, unregulated electrical discharge that occurs within the brain's cortical gray matter and transiently interrupts motor brain function.
- The term lumbago refers to lower back pain. The term sciatica refers to the pain along the sciatic nerve. It usually results from compression of nerve roots in the lower back. Common causes include intervertebral disc herniation, osteophytes and narrowing of spinal canal (spinal stenosis). Symptoms include pain radiating from lower back or buttocks to the lower limb. Diagnosis sometimes, involves MRI or CT scan.
- The term brachial neuralgia refers to a characteristic clinical syndrome of abrupt, severe pain in the shoulder and upper arm, that is followed several weeks later by muscle weakness and profound muscle wasting. It is also known as brachial neuritis, brachial amyotrophy. This is often associated with preceding viral illness, vaccination or other as brachial neuritis, brachial amyotrophy. This is often associated with preceding viral illness, vaccination or other immunological event. Affected muscles are most commonly supplied from C5/C6 segmental level, suggesting the involvement of upper trunk of brachial plexus, such as the long thoracic, axillary, suprascapular and anterior interosseus nerves.
- Cervical spondylosis is osteoarthritis of the cervical spine causing stenosis of the canal and sometimes cervical

myelopathy due to encroachment of bony osteoarthritic growths (osteophytes) on the lower cervical spinal cord, sometimes with involvement of lower cervical nerve roots (radiculomyelopathy).

- The term lumbar spondylosis is a degenerative disorder of the lumbo-sacral spine giving rise to disc degeneration and osteophytes formation, leading to spinal deformity, low back pain radiating down one or both lower limbs and features of lumbar radiculopathy.

Self-Assessment: Questions for Reviews

Short essay questions

- Explain Parkinson's disease
- Describe CVA (cerebrovascular accident) or stroke
- Explain Bell's palsy
- Describe MND (motor neuron disease)
- Explain transverse myelitis
- Describe organic epilepsy
- Explain sciatica syndrome
- Describe brachial neuralgia
- Describe cervical spondylosis
- Describe lumbar spondylosis

MCQs

1. Which of the following cranial nerve is involved in Bell's palsy?
 - a. Oculomotor nerve;
 - b. Facial nerve;
 - c. Glossopharyngeal nerve;
 - d. Vagus nerve

Answer: (b)

2. Which of the following is not the feature of Bell's palsy?
 - a. Obliteration of wrinkles of forehead;
 - b. Inability to close the affected eye;
 - c. Deviation of the mouth to the affected side of the face;
 - d. Deviation of the mouth to the sound side of the face

Answer: (c)

3. Which part of the body is afflicted in analysing Bell's phenomenon in Bell's palsy?
 - a. Tongue;
 - b. Nose;
 - c. Eye;
 - d. Forehead

Answer: (c)

4. Which of the following is not the feature of Parkinson's disease?
 - a. Bradykinesia;
 - b. Tremors;
 - c. Rigidity;
 - d. Hypotonia

Answer: (d)

5. When an individual is having the affliction of L1 and L2 intervertebral disc herniation, the person may have a radiation of pain
 - a. Anterior side of the lower limb;
 - b. Lateral side of the lower limb;
 - c. Posterior side of the lower limb;
 - d. Any or all of the above

Answer: (a)

6. Amyotrophic lateral sclerosis (ALS) is a variety of
 - a. Cerebrovascular accident;
 - b. Motor neuron disease;
 - c. Cervical spondylosis;
 - d. Transverse myelitis

Answer: (b)

7. Straight leg raising (SLR) test is performed to confirm the diagnosis of
 - a. Cerebrovascular accident;
 - b. Parkinson's disease;
 - c. Transverse myelitis;
 - d. Sciatica

Answer: (d)

8. Glabellar tap sign is performed to make the diagnosis of
 - a. Cerebrovascular accident;
 - b. Parkinson's disease;
 - c. Cervical spondylosis;
 - d. Transverse myelitis

Answer: (b)

9. Clasp knife spasticity is confirmatory which of the following disease?
 - a. Lumbar spondylosis;
 - b. Cerebrovascular accident;
 - c. Sciatica syndrome;
 - d. Transverse myelitis

Answer: (b)

10. Cogwheel rigidity is the sign observed in which of the following disease?
 - a. Lumbar spondylosis;
 - b. Cerebrovascular accident;
 - c. Parkinson's disease;
 - d. Transverse myelitis

Answer: (c)

Chapter resources

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